

GlobalTech Solutions -- System Architecture Specification

Version 2.4 | January 2025 | Confidential

1. Introduction

This document describes the architecture for GlobalTech's next-generation distributed data platform. The platform handles ingestion, processing, and querying of petabyte-scale datasets across three geographic regions.

1.1 Purpose

The primary purpose of this specification is to define the technical boundaries, component responsibilities, and integration contracts for all engineering teams contributing to the v2.4 release.

1.2 Scope

This document covers the following system components:

- Ingestion Layer: Apache Kafka + custom Flink pipelines
- Storage Layer: Delta Lake on S3 with Iceberg metadata catalog
- Query Layer: Trino cluster with adaptive query execution
- API Layer: GraphQL gateway with DataLoader batching
- Observability: OpenTelemetry + Prometheus + Grafana stack

2. Stakeholders and Responsibilities

The following individuals are the primary decision-makers and owners:

Alice Yamamoto -- Chief Architect, overall design authority
Ben Okafor -- Engineering Lead, ingestion and storage layers
Carmen Delgado -- Engineering Lead, query and API layers
David Park -- Security Lead, threat model and compliance
Elena Voroshilova -- Infrastructure Lead, Kubernetes and networking

2.1 Decision Log

The architecture review board meets weekly. Decisions require quorum of three senior engineers. All architectural decisions are recorded in the Architecture Decision Records (ADR) repository.

3. Architecture Decisions

ADR-001: Adopt Delta Lake for table format.

Decided by: Alice Yamamoto, approved by architecture board.

Rationale: ACID transactions, schema evolution, time travel queries.

Status: Accepted.

ADR-002: Replace custom message broker with Apache Kafka.

Decided by: Ben Okafor.

Rationale: Mature ecosystem, exactly-once semantics, compaction.

Status: Accepted.

ADR-003: Use GraphQL for the external API layer.

Decided by: Carmen Delgado.

Rationale: Type-safe contracts, introspection, client-driven queries.

Status: Accepted.

4. Component Specifications

4.1 Ingestion Layer

The ingestion layer receives events from three source systems: CRM, ERP, and IoT sensor network. Events are validated against Avro schemas stored in the Schema Registry before being forwarded downstream.

Throughput target: 500,000 events per second at p99 < 50ms latency.

4.2 Storage Layer

Raw events are written to Bronze Delta tables. The daily compaction job merges Bronze into Silver (cleaned) and Silver into Gold (aggregated).

Retention policy: Bronze 7 days, Silver 90 days, Gold unlimited.

4.3 Query Layer

Trino is deployed as a 24-node cluster: 4 coordinators and 20 workers.

Adaptive query execution is enabled. Result caching uses Alluxio.

SLA: 95th percentile query latency < 2 seconds for dashboard queries.

5. Security

All inter-service communication uses mTLS. API authentication uses JWTs issued by the internal identity provider (Keycloak). Data at rest is encrypted with AWS KMS. David Park owns the threat model review cycle.